

IDP Login – JMeter Performance Testing Capability & Limitations

What Can Be Done Using JMeter:

- We have a **working IDP login JMeter test suite** that can successfully simulate:
 - Basic login form submissions with valid credentials.
 - Hitting the **authentication server endpoints**.
 - **Increasing virtual users** for load and stress testing.
 - Simulating redirects handled at the HTTP level (e.g., 302/307).
 - Basic **session management**, including cookies and headers.
 - Partial CS page handling that behaves like static redirects or non-scripted responses.

What Cannot Be Done Using JMeter (Limitations):

Area	Limitation	Explanation
JavaScript Execution	Not Supported	JMeter is not a browser and does not execute client-side JavaScript . Any dynamic token generation or UI logic controlled via JS cannot be simulated.
AJAX/XHR Requests	Manual Setup Required	JMeter cannot auto-discover background AJAX/XHR calls triggered via frontend events. These must be manually identified and scripted.
Dynamic DOM & UI Events	Not Replicable	UI-driven logic (e.g., JS-based dropdown updates, event listeners, click handlers) won't work in JMeter , as it lacks a real browser engine.
CAPTCHA & MFA	Not Automatable	CAPTCHA or MFA flows requiring user input or SMS/email validation cannot be automated in JMeter unless test bypass logic is available.
Servlet Dispatch / CSPage Script Handling	Not Fully Replicable	Servlet-like behavior triggered by JavaScript or CSPage-based dynamic elements will not execute , as JMeter cannot mimic those flows without backend insight.
LocalStorage /SessionStorage	No Support	JMeter does not support reading or writing browser-based localStorage or sessionStorage where tokens or flags might be stored.
Full SSO Flow (IDP-initiated)	Partially Supported	Only achievable if all redirects and tokens are traceable via HTTP headers. Any JS-based injection will fail.

Summary of What Can Be Achieved

Capability	Status
Simulate login form submission	Working
Increase users and ramp up load	Working
Hit actual IDP authentication endpoints	Working
Manage cookies and headers	Working
Capture network request/response chain	Partially (manual correlation needed)
Handle JavaScript execution for token/captcha	Not Supported
Full WebAuthn / MFA validation	Not Supported
Session chaining across JS-driven flows	Not Reliable

Suggested Alternatives (if needed):

- For flows relying on JavaScript, DOM manipulation, or visual interaction:
 - Consider **JMeter + Selenium Hybrid scripts** using Taurus.
 - Explore tools like **Tricentis NeoLoad** or **Playwright/Gatling** if browser-like simulation is critical.
 - Request **backend bypass or test endpoint** for flows like CAPTCHA/MFA for test environments.